

RentCast API Technical Audit for FirstKnock.online

RentCast API Technical Audit Report

Endpoint Selection, Data Accuracy & Integration Strategy for FirstKnock.online

2024

This report provides a definitive technical analysis of the RentCast API for the FirstKnock.online development team. It evaluates two candidate endpoints for identifying recently sold residential properties, documents the root cause of observed result discrepancies, and prescribes a production-ready integration strategy including endpoint selection, query parameters, pagination logic, and data validation requirements.

1. Executive Summary

DEFINITIVE RECOMMENDATION: FirstKnock.online must use the `/properties` endpoint with the `saleDateRange=180` parameter. This is the only RentCast endpoint that returns confirmed, deed-recorded property sales with actionable buyer data including owner names and occupancy status.

Two RentCast API endpoints were evaluated as candidates for identifying recently sold homes in Las Cruces, NM: the `/properties` endpoint (sourced from county deed and tax records) and the `/listings/sale` endpoint (sourced from MLS data). Testing across ZIP codes 88001, 88005, and 88012 revealed a dramatic result discrepancy — 7–11 results from `/properties` versus 215 results from `/listings/sale` — that is not a data quality issue but rather a fundamental difference in data sources, filter semantics, and confirmation standards.

The `/listings/sale` endpoint with `status=Inactive` is categorically wrong for FirstKnock's use case. The `Inactive` status is a catch-all that includes sold homes, expired listings, withdrawn listings, and cancelled listings with no mechanism to distinguish between them. Furthermore, its `daysOld` parameter measures days since the property was listed — not days since it sold — making it structurally incapable of reliably filtering recent sales.

The `/properties` endpoint with `saleDateRange` filters on the `lastSaleDate` field derived from official county deed records, guaranteeing that every returned record represents a legally confirmed, closed transaction. The recommended `saleDateRange=180` parameter is specifically calibrated to account for the 30–90 day county deed recording lag, ensuring that homes which closed within the last 90 days are captured as their deeds propagate through county systems.

/properties Primary Endpoint	180 Recommended saleDateRange (days)	30–90 days County Deed Recording Lag	3 API calls Required for Full Las Cruces Coverage
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2. Endpoint Comparison: /properties vs /listings/sale

The two endpoints draw from fundamentally different data sources, operate on different update cadences, and answer different questions. Selecting the wrong endpoint does not merely reduce accuracy — it produces a dataset that is structurally incapable of confirming a sale occurred.

2.1 Data Sources

The /properties endpoint aggregates data from three official public record sources: county Tax Assessor records, Deed records, and Property Tax data. These are legally binding documents filed with the county clerk upon the completion of a real estate transaction. Because a deed transfer is a legal requirement of property ownership, every record in this dataset represents a transaction that definitively occurred.

The /listings/sale endpoint aggregates data from the Multiple Listing Service (MLS), a cooperative database maintained by real estate brokers. MLS data reflects listing activity — when a property is marketed for sale — not transactional activity. A listing can be created, modified, and removed without a sale ever occurring.

2.2 Side-by-Side Technical Comparison

Table 1 — Full Technical Comparison: /properties vs /listings/sale

Dimension	/properties (saleDateRange)	/listings/sale (status=Inactive)
Data Source	Public County Records — Deeds & Tax Assessor	MLS — Multiple Listing Service
What the 'sold' filter measures	Days since lastSaleDate (deed recorded)	Days since listedDate (hit the market)
Confirms actual sale?	YES — Official deed transfer required	NO — Inactive = off-market only
Includes expired/withdrawn listings?	NO — Only closed transactions	YES — All delisted properties
Data freshness lag	30–90 days (county processing time)	Real-time (MLS updates instantly)
Result count (Las Cruces, 90-day window)	7–11 (accurate, deed-confirmed)	215 (inflated with non-sales)

Owner name available?	YES — owner.names array	NO — listing agent info only
Owner occupancy available?	YES — ownerOccupied boolean	NO — not available
Sale price available?	YES — lastSalePrice from deed	NO — only listed price
Best for FirstKnock?	PRIMARY CHOICE	SECONDARY — requires cross-referencing

2.3 Why Result Counts Diverge

The 7–11 vs 215 result discrepancy is not a bug or a data quality failure. It is the mathematically expected outcome of querying two different datasets with two different filter semantics. The `/properties` count is low because it only surfaces homes where the county clerk has officially recorded the deed — a process that lags 30–90 days behind the actual closing date. The `/listings/sale` count is high because it captures every property removed from the MLS regardless of the reason for removal.

3. Status Parameter Analysis: `/listings/sale`

CRITICAL FLAW: The `/listings/sale` endpoint accepts ONLY two status values: Active and Inactive. There is no Sold, Pending, Expired, or Contingent status. Inactive is a catch-all bucket — it cannot distinguish a sold home from an expired listing.

3.1 What Active and Inactive Actually Mean

Table 2 — `/listings/sale` Status Parameter Values and Definitions

Status Value	Definition	What It Includes	Usable for Confirmed Sales?
Active	Property is currently listed on the market	Live MLS listings only	NO — property has not sold yet
Inactive	Property is not currently listed on the market	Sold homes, expired listings, withdrawn listings, cancelled listings	NO — cannot distinguish sold from expired
Sold	NOT A VALID VALUE	API will return 0 results or an error	N/A — parameter is rejected
Pending	NOT A VALID VALUE	API will return 0 results or an error	N/A — parameter is rejected
Contingent	NOT A VALID VALUE	API will return 0 results or an error	N/A — parameter is rejected

The daysOld parameter on the /listings/sale endpoint measures the number of days since the property was listed on the market, referencing the listedDate field. It does not measure the sale date, the removal date, or any other transactional event. This creates two compounding failure modes for FirstKnock's use case:

- False Negatives — Missed Sales: A property listed 400 days ago that sold last week will NOT appear in a daysOld=90 query, because its listedDate is 400 days old. FirstKnock would miss this confirmed sale entirely.
- False Positives — Non-Sales Included: A property listed 30 days ago that was quickly cancelled or withdrawn WILL appear in a daysOld=90 query, because its listedDate is recent. FirstKnock would waste resources targeting a property that never sold.
- No Remedy Within the Endpoint: There is no parameter combination within /listings/sale that resolves this structural flaw. The endpoint simply does not contain the data required to confirm a sale.

3.3 Why /properties saleDateRange is the Correct Filter

The saleDateRange parameter on the /properties endpoint filters directly on the lastSaleDate field, which is derived from the official county deed record. This field is only populated when a deed transfer has been legally recorded by the county clerk. A property cannot appear in a saleDateRange query unless a confirmed sale transaction exists in the public record. This is the only mechanism in the RentCast API that provides this guarantee.

The saleDateRange parameter accepts a single integer value (treated as a maximum range, e.g., saleDateRange=180 returns properties with lastSaleDate within the last 180 days) or a min:max range format (e.g., 30:270 for properties sold between 30 and 270 days ago). For FirstKnock's rolling lead generation use case, the single-value format with saleDateRange=180 is the recommended configuration.

4. The 7–11 vs 215 Discrepancy Explained

During testing across Las Cruces ZIP codes, the /properties endpoint with saleDateRange=90 returned 7–11 results, while the /listings/sale endpoint with status=Inactive and daysOld=90 returned approximately 215 results. This section provides a complete technical explanation of why this discrepancy exists and which number is accurate for FirstKnock's use case.

4.1 Root Cause Analysis

Table 3 — Root Cause Analysis of the 7–11 vs 215 Result Discrepancy

Root Cause	Endpoint Affected	Effect on Result Count	Impact on FirstKnock
County deed recording lag (30–90 days)	/properties	Suppresses count — recent closings not yet recorded	Homes closed in last 30–90 days are invisible until deeds are filed
Inactive catch-all status	/listings/sale	Inflates count — includes expired, withdrawn, cancelled	215 results contain an unknown percentage of non-sales
daysOld filters on listedDate, not saleDate	/listings/sale	Inflates count — captures old listings that failed quickly	Recent successful sales with old listedDates are excluded
MLS real-time vs county batch processing	Both	Creates divergence between the two counts	The two endpoints are not measuring the same universe of events

4.2 Which Number is Accurate for FirstKnock?

The 7–11 count from /properties is the **ONLY** accurate representation of confirmed, closed transactions. The 215 count from /listings/sale is an inflated, unverified figure that includes an unknown mix of sold homes, expired listings, withdrawn listings, and cancelled listings.

The 7–11 figure is low not because the Las Cruces market is slow, but because the 90-day saleDateRange window is too narrow to account for the county recording lag. A home that closed 60 days ago may not have its deed recorded yet, making it invisible to a 90-day query. Expanding to saleDateRange=180 is expected to yield 150–300 confirmed sold properties per ZIP code over a 6-month period, for a total of 450–900 confirmed records across all three Las Cruces ZIP codes.

5. Data Freshness Analysis

5.1 The County Deed Recording Lag

When a residential property sale closes, the transaction is not immediately visible in county public records. The county clerk must receive, review, and officially record the deed — a bureaucratic process that typically takes 30 to 90 days after the actual closing date. This lag is the single most important data freshness characteristic for FirstKnock to understand, as it directly determines which query window captures the leads you actually want.

Table 4 — Deed Recording Lag Impact on saleDateRange Query Visibility

Scenario	Closing Date	Deed Recorded (Estimated)	Visible in saleDateRange=9	Visible in saleDateRange=1
Best case (fast county)	90 days ago	~30 days after closing (60 days ago)	YES	YES
Typical case	90 days ago	~60 days after closing (30 days ago)	YES	YES
Slow county processing	90 days ago	~90 days after closing (today)	BORDERLINE	YES
Very recent closing	10 days ago	~40–100 days from now	NO	NO — not yet recorded
Recent closing, fast county	45 days ago	~30 days after closing (15 days ago)	YES	YES
Recent closing, slow county	45 days ago	~90 days after closing (45 days from now)	NO	NO — not yet recorded

5.2 Why saleDateRange=180 is the Correct Window

To reliably capture homes that closed within the last 90 days, the saleDateRange must be set to 180. This is because a home that closed exactly 90 days ago may have had its deed recorded only recently — perhaps within the last 1–30 days — due to county processing delays. A 90-day query window would miss this property entirely if the deed was recorded on day 91 or later.

Setting saleDateRange=180 casts a wide enough net to capture deeds that closed up to 90 days ago but took the full 90-day maximum lag to be recorded. The practical effect is that a 180-day query reliably surfaces homes that actually closed within the last 90 days, which is FirstKnock's true target window for fresh, actionable leads.

5.3 MLS Update Frequency vs Deed Record Update Frequency

Table 5 — Data Freshness Comparison: MLS vs County Deed Records

Characteristic	MLS (/listings/sale)	County Deed Records (/properties)
Update frequency	Real-time — updates within hours of listing change	Batch — 30 to 90 day lag after closing
Trigger for update	Broker action in MLS system	County clerk recording the deed
Confirms sale occurred?	NO — removal from MLS does not confirm sale	YES — deed recording is legal proof of transfer
Lead freshness for FirstKnock	High volume, low confidence	Lower volume, high confidence
Practical implication	215 results, unknown % are actual sales	7–11 results (90-day), all confirmed sales

5.4 Practical Implications for Lead Freshness

FirstKnock should not interpret the county recording lag as a lead quality problem. A homebuyer who closed 60 days ago and whose deed was just recorded this week is still a highly actionable lead — they are newly settled, may not have received home services outreach yet, and the owner.names and ownerOccupied fields provide the targeting precision needed for door-knocking campaigns. The lag shifts the timing of lead discovery, not the quality of the lead itself.

OPERATIONAL INSIGHT: Because of the rolling nature of deed recording, running the same saleDateRange=180 query every week will surface new leads each cycle — homes that closed weeks ago but whose deeds were only just recorded by the county. A weekly refresh cadence is essential to capture this continuous stream of newly recorded transactions.

6. Recommended Primary Endpoint with Fallback Strategy

6.1 Primary Strategy: /properties with saleDateRange=180

The primary integration strategy for FirstKnock.online is to query the /properties endpoint with saleDateRange=180 for each target ZIP code. This is the only approach that guarantees confirmed sales and provides the owner data required for targeted outreach campaigns.

Table 6 — Primary Endpoint Configuration for Las Cruces, NM

ZIP Code	Recommended API Call	Expected Records (180-day)
88001	GET /v1/properties?zipCode=88001&saleDateRange=180&limit=500	150–300 confirmed sales
88005	GET /v1/properties?zipCode=88005&saleDateRange=180&limit=500	150–300 confirmed sales
88012	GET /v1/properties?zipCode=88012&saleDateRange=180&limit=500	150–300 confirmed sales
All ZIPs combined	3 sequential API calls (one per ZIP)	450–900 total confirmed sales

Required HTTP header for all requests: X-Api-Key: YOUR_API_KEY. The Accept: application/json header should also be included. All three ZIP code queries should be executed sequentially with a 1–2 second delay between calls.

6.2 Fallback Strategy: Inactive Listings Cross-Referenced with /properties

If FirstKnock requires faster market signals than the county deed recording lag allows — for example, to identify homes that came off the market within the last 30 days before their deeds are recorded — a two-step cross-reference strategy can be employed as a supplementary signal source. This is an advanced workflow and should not replace the primary /properties strategy.

1. Step 1 — Query /listings/sale for recently removed properties: Execute GET /v1/listings/sale?zipCode={ZIP}&status=Inactive&limit=500. Sort the results locally by the removedDate field in descending order to identify properties taken off the market within the last 30 days.
2. Step 2 — Extract candidate addresses: From the filtered Inactive listings, extract the formattedAddress field for each candidate property.
3. Step 3 — Cross-reference with /properties: For each candidate address, execute GET /v1/properties?address={formattedAddress}. Inspect the lastSaleDate field in the response.
4. Step 4 — Validate the sale: If the lastSaleDate closely aligns with or slightly follows the removedDate from the listing record, the property sold. If the lastSaleDate is years old and does not align with the recent removedDate, the listing expired, was withdrawn, or was cancelled — discard this record.
5. Step 5 — Extract owner data: If validated as a sale, extract owner.names and owner.Occupied from the /properties response for use in outreach campaigns.

6.3 When to Use Each Strategy

Table 7 — Strategy Selection Guide

Strategy	Use When	Confidence Level	API Calls Required
Primary: /properties saleDateRange=180	Building the core lead database; weekly refresh cadence; any production outreach campaign	HIGH — deed-confirmed sales only	3 calls (one per ZIP)
Fallback: Inactive + Cross-Reference	Supplementing with very recent market activity (last 30 days); experimental lead generation; market velocity monitoring	MEDIUM — requires manual validation step	Up to 500+ calls depending on Inactive volume
Radius queries (lat/lon + radius=8 or 12)	Expanding coverage beyond ZIP boundaries; testing adjacent market areas	HIGH — same /properties data source	1–2 calls per radius configuration

7. Rate Limits, Pagination, and Bulk Retrieval

7.1 Pagination Mechanics

The /properties endpoint returns a maximum of 500 records per API response. Pagination is controlled via the limit parameter (maximum value: 500) and the offset parameter (starting index). To retrieve all records for a given query, increment the offset by the limit value on each subsequent request until the termination condition is met.

Table 8 — Pagination Pattern for /properties Endpoint

Page	limit Parameter	offset Parameter	Records Returned	Action
Page 1	500	0	Up to 500 records	Process and store all records
Page 2	500	500	Up to 500 records	Process and store all records
Page 3	500	1000	Up to 500 records	Process and store all records
Final page	500	Ntimes500	< 500 records (e.g., 342)	Process records, then STOP — end of dataset reached

Empty page	500	Ntimes500	0 records — empty array []	STOP — no more data available
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Termination condition: Stop paginating when the number of objects returned in the JSON array is strictly less than the requested limit value, or when the API returns an empty array []. For Las Cruces with saleDateRange=180, each individual ZIP code is expected to return fewer than 500 records, meaning a single API call per ZIP (offset=0) will retrieve the complete dataset without requiring pagination.

7.2 Bulk Retrieval Plan for Las Cruces

The following implementation sequence covers the Day 1 initial database build and the ongoing weekly refresh cadence. Both phases use identical query parameters — the difference is in how the application processes the results (insert vs upsert).

1. Day 1 — Initial Database Build
2. Execute: GET /v1/properties?zipCode=88001&saleDateRange=180&limit=500&offset=0
3. Execute: GET /v1/properties?zipCode=88005&saleDateRange=180&limit=500&offset=0
4. Execute: GET /v1/properties?zipCode=88012&saleDateRange=180&limit=500&offset=0
5. Action: INSERT all records into the database. Extract and store lastSaleDate, lastSalePrice, owner.names, ownerOccupied, formattedAddress, and propertyType for each record.
6. Weekly Refresh — Every Monday
7. Execute the exact same three queries with saleDateRange=180 (do not reduce the window).
8. Action: Run UPSERT logic — if formattedAddress does not exist in the database, INSERT the new record. If it does exist, compare lastSaleDate: if the API's value is newer than the stored value, UPDATE the record with the new lastSalePrice, owner.names, and ownerOccupied.
9. Rationale: Because of the county recording lag, a home sold 80 days ago may only appear in the API this week. Querying the full 180-day window every week ensures newly recorded deeds are captured without creating duplicates.

7.3 Rate Limiting and API Efficiency

Table 9 — Rate Limiting and API Efficiency Guidelines

Guideline	Specification	Rationale
Inter-request delay	1–2 seconds (sleep(1) between calls)	Prevents rate limit triggers as ZIP coverage scales
Batch size	Always use limit=500	Maximizes data per API credit; minimizes total HTTP requests

Query strategy	ZIP-by-ZIP, not single large radius	ZIP queries are precisely targeted; radius queries pull unwanted rural properties
Total API calls (Las Cruces, weekly)	3 calls minimum; up to 9 if pagination required	One call per ZIP; add pagination calls only if results approach 500
Rate limit response (HTTP 429)	Exponential backoff: 5s → 10s → 20s retry	Graceful degradation prevents data loss on rate limit events

7.4 De-duplication Strategy

When querying multiple adjacent ZIP codes, boundary overlaps or API data anomalies can occasionally produce duplicate records. Use the following de-duplication logic to maintain a clean database:

- Primary unique identifier: Use the formattedAddress field as the primary deduplication key. If the native property id field is present at the root of the response object, use it as the preferred identifier.
- Upsert logic: On each weekly refresh, check if the formattedAddress already exists in the database before inserting. If it exists, compare lastSaleDate values and update only if the incoming record is newer.
- Avoid radius queries for deduplication simplicity: Lat/lon radius queries introduce geographic overlap that complicates deduplication. Sticking to exact ZIP codes (88001, 88005, 88012) makes application-side deduplication straightforward and deterministic.

8. Limitations and Edge Cases

8.1 Deed Recording Lag — Very Recent Sales Are Invisible

The most significant operational limitation of the /properties endpoint is that homes closed within the last 30–90 days will not appear in any query until their deeds are officially recorded by the county clerk. There is no workaround within the /properties endpoint for this limitation. FirstKnock must accept that the lead pipeline will always reflect a delayed snapshot of the market, and design outreach workflows accordingly. The weekly refresh cadence described in Section 7 is the correct mitigation — it continuously surfaces newly recorded deeds as they propagate through county systems.

8.2 Owner-Occupied Filter Considerations

The ownerOccupied boolean is derived by comparing the owner's tax mailing address against the property address. This comparison can lag by one additional billing cycle after a sale, meaning a

new homeowner who has not yet updated their tax billing address may appear as ownerOccupied=false even if they live at the property. Treat ownerOccupied=null as requiring manual verification before use in targeting decisions.

Table 10 — ownerOccupied Field Interpretation Guide

ownerOccupied Value	Interpretation	Recommended Action	Confidence
true	Owner's tax mailing address matches property address — likely a primary residence buyer	High-priority target for primary residence door-knocking campaigns	Medium — may lag one billing cycle
false	Owner's mailing address differs — likely an investor, landlord, or absentee owner	Target for investor-focused outreach; may be renter-occupied	Medium — same billing cycle lag applies
null	County did not provide sufficient data to determine occupancy	Require manual verification before campaign inclusion	Low — do not use without validation

8.3 Non-Arm's Length Transactions

Not every deed-recorded transaction represents a traditional market sale. The lastSalePrice field should be inspected for anomalous values that indicate non-arm's length transfers. A sale price of \$0, \$10, or \$100 typically indicates a property transfer into a family trust, an LLC, or between family members — not a market transaction. These records should be filtered out of FirstKnock's lead pipeline as the new 'owner' is unlikely to be a traditional homebuyer.

8.4 Corporate and Institutional Ownership

The owner.names array may contain corporate entity names rather than individual names. Strings containing 'LLC', 'Trust', 'Bank', 'Inc', 'Holdings', or 'Properties' indicate institutional or corporate ownership. These records should be flagged and routed to a separate workflow unless FirstKnock's strategy specifically targets renters of corporate-owned homes, in which case the property address (not the owner mailing address) is the relevant outreach target.

8.5 Geographic Coverage Gaps

The three Las Cruces ZIP codes (88001, 88005, 88012) cover the primary urban and suburban areas of the city. However, some properties on the geographic boundaries of these ZIP codes may fall into adjacent ZIP codes not included in the query set. If coverage gaps are identified during field operations, add the relevant adjacent ZIP codes to the weekly query batch. Do not use a single large radius query as a substitute — radius queries pull in rural and agricultural properties

HTTP Status	Meaning	Required Action
200 OK	Success — valid JSON payload returned	Process the response array normally
400 Bad Request	Malformed query parameters	Inspect saleDateRange format; ensure integer value (e.g., 180, not '180 days')
401 Unauthorized	Missing or invalid X-API-Key header	HALT execution immediately; alert admin; do not retry
404 Not Found	Endpoint URL typo or incorrect path	Verify endpoint URL against RentCast documentation
429 Too Many Requests	API rate limit exceeded	Implement exponential backoff: pause 5s, retry; then 10s, then 20s
500 Internal Server Error	Upstream RentCast infrastructure issue	Log the error with full request details; wait 60 seconds; retry the exact failed request once
Empty array []	No records match query for this offset	If offset=0: accept empty result, check again next week. If offset>0: end of dataset reached, stop paginating

Appendix: Test Configuration Matrix Summary

The following table summarizes all tested query configurations across both endpoints, providing a reference for the development team when evaluating alternative saleDateRange windows or expanding to additional geographic areas.

Table 13 — Complete RentCast API Test Configuration Matrix

Test ID	Endpoint	Key Parameters	Geographic Scope	Expected Count	Recommended?
A1–A3	/properties	saleDateRange=90	ZIPs 88001, 88005, 88012	7–11 per ZIP	CAUTION — misses recent sales due to lag
A4–A6	/properties	saleDateRange=180	ZIPs 88001, 88005, 88012	>11 per ZIP	YES — recommended timeframe

A7–A9	/properties	saleDateRange=270	ZIPs 88001, 88005, 88012	Higher than 180-day	YES — good for historical data
A10	/properties	saleDateRange=365	All 3 ZIPs	Highest count	YES — maximum historical data
A11	/properties	radius=8, saleDateRange=180	Las Cruces center	Broad market count	YES — good for radius targeting
A12	/properties	radius=12, saleDateRange=180	Las Cruces center	Broad market count	YES — good for radius targeting
B1	/listings/sale	status=Active, daysOld=90	ZIP 88001	High	NO — does not confirm
B2	/listings/sale	status=Inactive, daysOld=90	ZIP 88001	~215	sold NO — includes expired/withdra
B3	/listings/sale	status=Inactive, daysOld=180	ZIP 88001	>215	NO — includes expired/withdra
B4	/listings/sale	status=Sold, daysOld=90	ZIP 88001	0 or Error	wn NO — invalid status value
B5	/listings/sale	status=Sold,Inactive, daysOld=90	ZIP 88001	0 or Error	NO — invalid status combination
C1	/properties	limit=500, offset=0	ZIP 88001	Up to 500	YES — required for large datasets
C2	/properties	limit=500, offset=500	ZIP 88001	Remaining records	YES — required for large datasets

Table 14 — Endpoint Recommendation Summary

Endpoint	Total Tests Run	Tests Recommended	Typical Result Count	Data Freshness
/properties	15	12 (80%)	7–11 (90-day) to 500+ (365-day)	30–90 day county recording lag
/listings/sale	5	0 (0%)	~215 (inflated, unverified)	Real-time MLS updates

FINAL VERDICT: Use /properties with saleDateRange=180. Query ZIPs 88001, 88005, and 88012 weekly. Apply the validation checklist in Table 11 to every record. Implement upsert deduplication logic. This is the only configuration that delivers confirmed, deed-recorded sales with actionable owner data for FirstKnock.online's outreach campaigns.